

gen-Ori Project Documentation

Version: Current Research Proposal

Author: Hoàng Đức

Part I — Project Overview

1. Introduction

gen-Ori is a long-term research project that aims to build an AI-assisted origami design system capable of transforming human imagination into origami designs.

Rather than viewing origami generation as a problem of directly producing a crease pattern (CP) from a text prompt, gen-Ori follows the actual workflow used by human origami designers. The project is based on the observation that experienced designers rarely think directly in terms of crease patterns. Instead, they gradually transform an abstract idea into increasingly structured representations before eventually arriving at a foldable design.

The central vision of the project is therefore:

To close the gap between human imagination and origami execution by allowing people to describe what they imagine, while AI translates that imagination into representations that can interact with mathematical origami design methods and eventually become understandable folding instructions.

Instead of replacing existing origami mathematics, gen-Ori aims to bridge the communication gap between humans and those mathematical systems.

2. Overall Philosophy

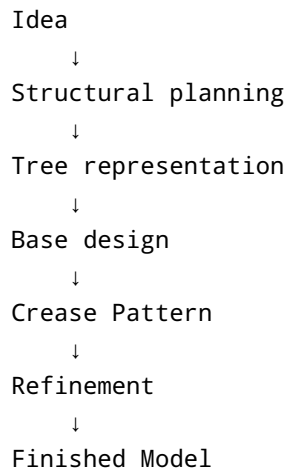
The project follows one fundamental principle:

AI should understand human intent and communicate with existing origami mathematics rather than replacing it.

Many current generative AI systems attempt to directly generate images, code, or other outputs from user prompts.

Origami design, however, is fundamentally different.

Professional origami designers usually follow a design pipeline similar to:

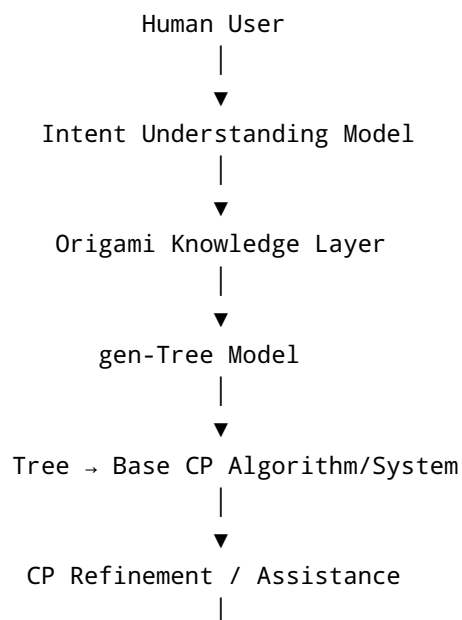


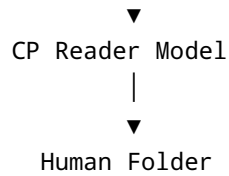
This pipeline contains several intermediate reasoning steps that are difficult to replace with a single end-to-end model.

Therefore, gen-Ori decomposes the problem into multiple specialized AI components, each responsible for one stage of the overall design process.

3. High-Level System Pipeline

The current architecture is shown below.





Each component has a clearly defined responsibility.

4. System Components

4.1 Intent Understanding Model

Purpose:

Understand what the user wants to create.

This model accepts user input in various forms, such as:

- Natural language
- Sketches
- Simple drawings
- Stick figures
- Other multimodal inputs

Its responsibility is **not** to generate origami.

Instead, it extracts semantic information from the user's idea and converts it into a structured representation describing the requested object and design intent.

4.2 Origami Knowledge Layer

Purpose:

Provide origami-specific knowledge that general language models do not possess.

The Knowledge Layer bridges human concepts and origami concepts by supplying information such as:

- object structures
- symmetry
- common origami bases
- design rules
- origami terminology

- relationships between structural components
-

4.3 gen-Tree Model

Purpose:

Generate an origami tree representation from the interpreted design requirements.

The tree acts as an intermediate structural representation between semantic understanding and mathematical origami algorithms.

Rather than reasoning about folds directly, the model reasons about structural allocation.

4.4 Tree-to-Base CP System

Purpose:

Convert the generated tree into a valid base crease pattern.

Instead of developing entirely new mathematical algorithms, gen-Ori intends to leverage existing origami research (e.g., TreeMaker-style approaches) to perform this conversion.

This allows the project to focus on AI-driven semantic reasoning while relying on proven mathematical methods for geometric optimization.

4.5 CP Refinement / Assistance

Purpose:

Assist in transforming a base crease pattern into a refined and more complete design.

Future work may include:

- improving proportions
 - suggesting modifications
 - comparing alternative designs
 - refining structural details
-

4.6 CP Reader

Purpose:

Translate crease patterns into understandable folding instructions.

A crease pattern is useful to experienced folders but difficult for beginners to interpret.

The CP Reader aims to convert crease patterns into human-readable folding steps.

5. Current Development Focus

The project currently concentrates on the first three components:

- Intent Understanding Model
- Origami Knowledge Layer
- gen-Tree Model

These three components together define how human ideas are converted into structured origami representations.

The remaining components will be developed in later stages.

Part II — Detailed Design of Current Research Components

1. Intent Understanding Model

1.1 Objective

The Intent Understanding Model is responsible for interpreting the user's request before any origami-specific reasoning occurs.

Its primary goal is to convert human intention into a structured semantic representation that can be used by later components. In practice, this means identifying what the user wants, what attributes are explicitly stated, and what information may need to be inferred. This stage is intentionally independent of origami mathematics.

1.2 Why Separate Intent Understanding?

A large language model is already capable of understanding natural language reasonably well.

However, directly asking it to generate origami structures mixes two different tasks:

1. Understanding language.
2. Designing origami.

1.3 Intent Model Pipeline

The Intent Understanding Model is designed as a small pipeline rather than a single monolithic step.

A typical flow is:

```
graph TD; A[User Input (natural language)] --> B[LLM]; B --> C[Structure Inference + Intent Extraction]; C --> D[Structured Output (Json format)];
```

A. Intent Extraction

After receiving the raw user input, the model identifies the user's explicit request.

This stage focuses on answering questions such as:

- What object is being requested?
- What style or pose is desired?
- What constraints are mentioned?
- What level of complexity is expected?
- What additional requirements are present?

This step captures the user's direct intention without yet making origami-specific decisions.

B. Structure Inference

Once the intent is identified, the model infers the likely physical structure of the requested object.

For example, if the user requests a cat, the model may infer that the object likely contains:

- head
- body
- four legs
- tail
- ears

This stage does not finalize the structure; it only proposes an initial structural interpretation that can later be refined by the Knowledge Layer.

2. Origami Knowledge Layer

2.1 Objective

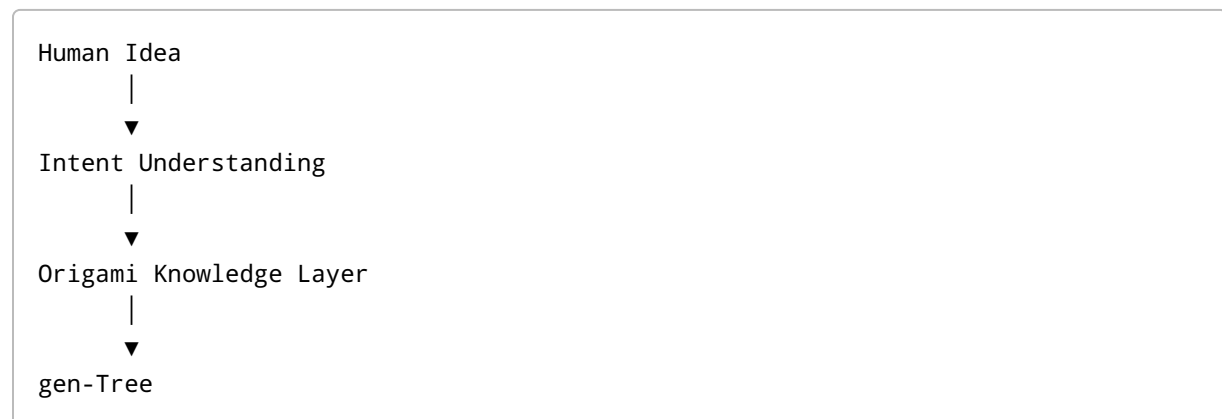
The Knowledge Layer serves as the domain intelligence of gen-Ori.

Large language models possess extensive general knowledge but relatively limited understanding of origami design principles.

The Knowledge Layer supplies the missing expertise.

Its purpose is to transform semantic concepts into origami-aware concepts.

2.2 Position Within the Pipeline



The Knowledge Layer sits between language understanding and structural generation.

2.3 Responsibilities

The Knowledge Layer performs several responsibilities.

A. Origami Knowledge

Provide factual information relevant to origami design.

Examples include:

- common bases
 - symmetry
 - folding techniques
 - structural conventions
 - flap allocation tendencies
-

B. Constraint Information

The Knowledge Layer may also provide constraints relevant to origami design, such as:

- symmetry requirements
 - required appendages
 - expected complexity
 - preferred structural layouts
-

2.4 Initial Implementation

The first prototype will use manually written structured files.

For example:

```
knowledge/  
  
  animals.json  
  
  bases.json  
  
  symmetry.json  
  
  techniques.json  
  
  design_rules.json
```

topology_difficulty.json

2.5 Example Knowledge Entry

An individual entry may resemble:

```
{
  "cat": {
    "category": "animal",
    "symmetry": "bilateral",
    "parts": [
      "head",
      "body",
      "four legs",
      "tail",
      "ears"
    ],
    "recommended_structures": [
      "quadruped"
    ]
  }
}
```

The exact schema is expected to evolve as development progresses.

2.6 Future Evolution

As the project grows, manually written files will become increasingly difficult to maintain.

The Knowledge Layer is therefore designed to evolve toward richer representations such as:

- vector databases
- embedding retrieval
- retrieval-augmented generation (RAG)
- knowledge graphs

The primary motivation is not storage efficiency but semantic reasoning.

Rather than relying solely on keyword matching, future versions should retrieve concepts based on meaning and relationships.

3. gen-Tree Model

3.1 Objective

The gen-Tree Model converts structured design information into an origami tree representation (a stick figure).

This tree serves as the interface between AI reasoning and mathematical origami algorithms.

Instead of deciding folds directly, the model decides how the object's structural components should be organized.

3.2 Why a Tree?

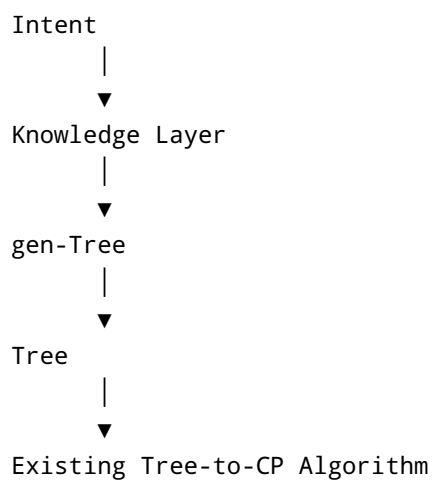
The tree representation is inspired by how many computational origami design methods represent flap allocation.

It also mirrors the reasoning process of many experienced origami designers.

Rather than thinking about individual folds first, designers often think about the arrangement of major structural components.

The tree therefore becomes an intermediate language shared between semantic reasoning and geometric algorithms.

3.3 Position Within the Pipeline



3.4 Current Scope

At the current stage of research, the objective is limited to generating the tree representation itself.

The project deliberately avoids introducing new mathematical methods for converting trees into crease patterns.

Instead, existing algorithms will be used for that purpose.

3.5 Parent Prediction Decoder

The current conceptual direction for the decoder is based on **parent prediction**.

Rather than generating the entire tree in one step, the model predicts how structural components connect by assigning a parent to each node.

Conceptually, the decoder receives a collection of structural components and predicts the hierarchical relationship among them.

For example, given components such as:

- head
- body
- tail
- left wing
- right wing

the decoder predicts which component each node should attach to, ultimately forming a valid tree.

This approach naturally produces hierarchical structures while remaining compatible with graph-based representations.

At this stage, parent prediction is considered a promising decoding strategy, but the detailed implementation and learning procedure remain future work.

3.6 Future Research

Several aspects of the gen-Tree Model remain active research topics, including:

- representation learning
- graph encoding
- decoding strategies
- training datasets

- evaluation methods
- integration with downstream Tree-to-CP algorithms

These topics are intentionally left open while the current prototype focuses on defining a clear and robust system architecture.

4. Current Status

The conceptual architecture of gen-Ori has largely stabilized around a modular pipeline in which semantic understanding, origami knowledge, and structural generation are treated as separate but cooperating components.

Current development is focused on designing and prototyping:

1. the Intent Understanding Model,
2. the Origami Knowledge Layer, and
3. the gen-Tree Model.

These components establish the foundation upon which future work—including crease pattern generation, refinement, and folding instruction generation—will be built.

The long-term objective remains unchanged:

To bridge the gap between human imagination and origami execution through AI-assisted semantic understanding, structured origami representations, and existing mathematical origami design algorithms.